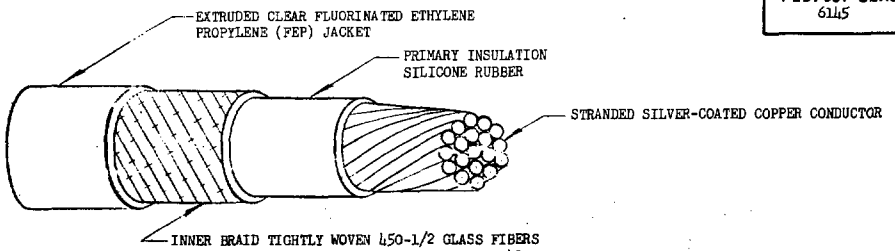
FED. SUP CLASS
6145




| PHYSICAL PROPERTIES OF PRIMARY INSULATION | |
|---|---|
| CHARACTERISTIC | REQUIREMENT |
| TENSILE STRENGTH MIN | BEFORE AGING 800 PSI<br>AFTER AGING 600 PSI |
| ELONGATION MIN | BEFORE AND AFTER AGING, 1-1/2 INCHES (2 INCH SPECIMEN STRETCHED TO 3-1/2 INCHES) |

PERFORMANCE DETAILS

| WIRE SIZE | ABRASION TEST: RESISTANCE INCHES OF TAPE MIN | ABRASION TEST: TENSION LOAD LBS | ABRASION TEST: WEIGHT SUPPORT BRACKET | ABRASION TEST: WEIGHT POUNDS | FLAWS TEST: MIN AC VOLTAGE 60 CYCLE RMS: PRIMARY INSULATION | FLAWS TEST: MIN AC VOLTAGE 60 CYCLE RMS: FINISHED WIRE | INSULATION AND SURFACE RESISTANCE: HUMIDITY RESISTANCE MEGOHMS PER 200 FT MIN | INSULATION AND SURFACE RESISTANCE: INSULATION RESISTANCE MEGOHMS PER 50 FT MIN | INSULATION AND SURFACE RESISTANCE: SURFACE RESISTANCE MEGOHM INCHES MIN | LIFE CYCLE AND COLD BEND: MANDREL DIA: LIFE CYCLE | LIFE CYCLE AND COLD BEND: MANDREL DIA: COLD BEND | LIFE CYCLE AND COLD BEND: TEST LOAD LIFE CYCLE COLD BEND LBS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 22 | 22 | 1 | A | 1 | 2,000 | 5,000 | 3,000 | 1,000 | 100 | 4-1/2 | 3 | .75 |
| 20 | | | | | | | | | | | | |
| 18 | | | | | | | | | | 6-1/2 | | 1.0 |
| 16 | 30 | 2 | | | | | | | | | | |
| 14 | 13 | | B | 3 | | | | | | | 6 | 3.0 |
| 12 | 17 | | | | 3,000 | | | | | | | |
| 10 | 20 | | | | | | | | | 10 | | |
| 8 | 25 | | C | 4.25 | 4,000 | | | | | | 10 | 6.0 |
| 6 | | | | | | | | | | | | |
| 4 | 33 | | | | | | | | | | | |

| FINISHED WIRE CONSTRUCTION | | | | | | | |
|---|---|---|---|---|---|---|---|
| DASH NO. | WIRE SIZE | NUMBER OF STRANDS | RESISTANCE AT 20°C (68°F) OHMS/1,000 FT MAX | FINISHED WIRE DIA NOM | | FEP JACKET THICKNESS NOM | | WEIGHT FINISHED WIRE LBS/1,000 FT MAX |
| -22 | 22 | 19 | 15.2 | .086 | ±.004 | .007 | +.002 -.001 | 7.1 |
| -20 | 20 | | 9.42 | .096 | | | | 9.0 |
| -18 | 18 | | 6.03 | .108 | | | | 12.1 |
| -16 | 16 | | 4.76 | .116 | | | | 14.3 |
| -14 | 14 | | 2.99 | .141 | | | | 21.5 |
| -12 | 12 | | 1.88 | .160 | ±.005 | .008 | | 30.5 |
| -10 | 10 | 49 | 1.16 | .194 | | | | 48.0 |
| -8 | 8 | 133 | .70 | .243 | ±.007 | .010 | ±.002 | 75.0 |
| -6 | 6 | | .436 | .292 | | | | 114.0 |
| -4 | 4 | | .274 | .357 | | | | 173.0 |

(C) REQUIREMENT: DURING AND FOLLOWING THE THERMAL SHOCK TEST, NONE OF THE INSULATION SHALL SHRINK BACK OR FLARE GREATER THAN .060 FOR WIRE SIZES 22-12 AND .125 FOR WIRE SIZES 10-4.

TWO 24-INCH SAMPLES SHALL BE TAKEN FROM EACH REEL OF FINISHED WIRE OR 25,000 FEET, WHICHEVER IS LESS. THE SAMPLES SHALL BE PLACED ON MANDRELS WITH WEIGHTS ATTACHED AS SPECIFIED FOR THE LIFE CYCLE TEST AND SUBJECTED TO 200° ±5° C FOR 20 HOURS. FOLLOWING THE AIR OFEN TEST, THE SAMPLES SHALL BE COOLED FOR ONE HOUR AND SUBJECTED TO THE BEND TESTS OF 4.6.4.3.2 AND THE DIELECTRIC TEST OF 4.6.4.3.3. DURING THESE TESTS, THE OUTER JACKET SHALL NOT SHRINK MORE THAN 1/8 INCH FROM EACH END.

WHEN PRINTING (AS SPECIFIED IN THE IDENTIFICATION OF PRODUCT REQUIREMENT OF 3.4) IS MARKED ON AN INNER BRAID IT SHALL BE VISIBLE THROUGH THE OUTER FEP JACKET.

DIMENSIONS IN INCHES.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

APPROVED 14 Feb 61 REVISED (A) 30 Jun 68 (B) 7 Mar 66 (C) 6 Jan 69

Review activities: USAF - 11 Navy - AS

This military standard is approved by the Department of the Air Force and the Naval Air Systems Command and is mandatory for use by these activities. All other military activities are required to employ this standard where suitable.

| P.A. USAF - 11<br>Other Cust<br>Navy - AS | TITLE<br>WIRE, ELECTRICAL - SILICONE INSULATED, COPPER, 600 VOLT, 200°C, FEP JACKET | MILITARY STANDARD<br>MS27110(ASG) |
|---|---|---|
| PROCUREMENT SPECIFICATION<br>MIL-W-8777 | SUPERSEDES: | SHEET 1 OF 1 |

DD FORM 1 SEP 67 672-1 (Limited coordination)
ASG use only

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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